High-Performance Multimodal Enterprise RAG Benchmark Document

1. Theoretical Framework & Quantum Formulations

In quantum mechanics, the time-dependent Schrödinger equation describes the wave function evolution: $$i\hbar \frac{\partial}{\partial t} \Psi(\mathbf{r}, t) = \left[-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{r}, t) \right] \Psi(\mathbf{r}, t)$$

where $\hbar$ is the reduced Planck constant, m represents particle mass, and $V(\mathbf{r}, t)$ denotes the potential field.

For relativistic systems, the Dirac Hamiltonian equation governs spin-1/2 fermionic fields: $$H_D = c \boldsymbol{\alpha} \cdot \mathbf{p} + \beta m c^2$$

The eigenvalues of the Hamiltonian correspond to the observable stationary energy states $E_n = n \hbar \omega$.

Table 1: Fundamental Physical Constants and Convergence Limits

Constant / Symbol	Value	Unit	Uncertainty (ppm)
Planck Constant (h)	6.62607015e-34	Joule-second (J*s)	
Speed of Light (c)	299792458	Meters per second (m/s)	Exact (defined)
Electron Mass (m_e)	9.1093837e-31	Kilograms (kg)	0.003 ppm
Boltzmann Constant (k_B)	1.380649e-23	Joules per Kelvin (J/K)	
Fine-Structure Const (alpha)	0.007297352569	Dimensionless	0.00015 ppm

Statistical convergence was verified across 10,000 Monte Carlo trials. Notice that fine-structure constant alpha demonstrates stability within 0.00015 ppm.

2. Financial Risk Metrics & Architectural Benchmarks

Quantitative options pricing utilizes the Black-Scholes partial differential equation:
$$\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + r S \frac{\partial V}{\partial S} - r V = 0$$

Here, S denotes asset spot price, σ represents annualized volatility, and r is the risk-free interest rate.

Table 2: Enterprise Hybrid RAG Performance Latency SLA Benchmarks

Pipeline Stage	Method / Engine	Mean Latency (ms)	SLA Target (ms)
Dense Vector Search	ChromaDB HNSW Cosine	42.5 ms	< 100 ms
Sparse Lexical Search	BM25Okapi Tokenized	18.3 ms	< 50 ms
Reciprocal Rank Fusion	RRF (k = 60)	2.1 ms	< 10 ms
Cross-Encoder Rerank	ms-marco-MiniLM-L-6-v2	115.4 ms	< 200 ms
End-to-End Retrieval	Hybrid + Rerank Total	178.3 ms	< 400 ms

Conclusion: The end-to-end hybrid retrieval and cross-encoder re-ranking stage operates at 178.3 ms, well within the strict production SLA ceiling of <400 ms.